

Using OLS Regression to Predict Median House Values in Philadelphia

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Introduction

This report examines the relationship between median house values and several neighborhood characteristics across Philadelphia's census block groups. The ability to predict median house values provide insight on the housing market for politicians, developers, and city planners that are working to reshape the city. However, it is not always clear on how to accomplish this due to a plethora of factors which determine property values.

To conduct this analysis, the following variables were used: the proportion of residents in a Philadelphian block group with at least a bachelor's degree (PCBACHMORE), the proportion of vacant housing units (PCTVACANT), the percent of housing units that are detached single-family houses (PCTSINGLES), the number of households in poverty (NBELPOV100), and the median household income (MEDHHINC). Previous studies in other locations, such as Taiwan, have used neighborhood characteristics such as median income, crime rates, school quality, proximity to amenities, and demographic (racial, ethnic, and age) composition (Chang & Lin, 2022). In the United States, demographic changes have largely accounted for house price growth between 1970 and 2010, with projected changes in the country's demographic factors leading to future house price growth (Gong & Yao, 2022). Following these papers, it is reasonable to assume that neighborhoods with high poverty often have low-income households, which can only afford cheaper housing, resulting in lower property values. In contrast, single-family homes are generally more desirable and linked to higher values. The vacancy rate is also a key indicator of demand; high vacancy rates suggest lower demand, leading to reduced prices. These four factors will be used to estimate median house values in Philadelphia's census blocks.

Methods

Data Cleaning

Within the 2000 Philadelphia Census block group level dataset *RegressionData.csv*, there were 1816 observations with the variables such as: POLY_ID (census block group ID); MEDHVAL (median value of all owner occupied housing units); PCBACHMORE (Proportion of residents in Block Group with at least a bachelor's degree); PCTVACANT (Proportion of housing units that are vacant); PCTSINGLES (Percent of housing units that are detached single family houses); NBELPOV100 (Number of households with incomes below 100% poverty level (i.e., number of households living in poverty); MEDHHINC (Median household income), etc. This data was clean by removing the following: block groups where population was under 40; block groups where there are no housing units; Block groups where the median house value is lower than \$10,000; One North Philadelphia block group which had a very high median house value (over \$800,000) and a very low median household income (less than \$8,000). Following the data cleaning, the data set is left with 1720 block groups.

Exploratory Data Analysis

An exploratory data analysis is conducted to observe any underlying patterns. This is done by examining summary statistics, distributions of variables, and correlations between the predictors. A correlation is a statistical measure describing the strength and direction of a relationship between two variables.

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Where:

- r : the Pearson correlation coefficient
- n : the total number of paired data point
- x_i : the i -th value of variable x
- y_i : the i -th value of variable y

- $\bar{x}$: the mean of all x values
- $\bar{y}$: the mean of all y values
- s_x : the standard deviation of the x values, calculated as $\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$
- s_y : standard deviation of the y values $\sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$

Correlation (r) does not depend on units of measurement of x and y . r is referred to as the Pearson's product moment correlation, or simply Pearson correlation. A correlation can be either positive or negative, ranging from 1 to -1. A r value of 1 would indicate a perfectly positive correlation while a r value of -1 would indicate a perfectly negative correlation. A positive correlation indicates that as the values of one variable increase, so do the values of the other variable. A negative correlation indicates that as the values of one variable increases, the values of the other variable decreases. If there is a correlation of 0, then there is no linear relationship between the variables. The strength of the correlation can be determined by the range below.

- $0 \leq |r| \leq 0.5$ is a weak correlation
- $0.8 \leq |r| \leq 1$ indicates a strong correlation
- An r value between 0.5 and 0.8 indicates a moderate correlation.

Multiple Regression Analysis

Linear Regression

Regression is a statistical method used to examine the relationship between a variable of interest, the dependent variable, and one or more predictor variables. It assumes a linear relationship between the variables and fits a line, plane or hyperplane, depending on the number of variables, through the data points, aiming to quantify the nature of the relationship between

the predictors and outcome variable. This includes the strength as well as the direction (positive, negative, zero) of the relationship. Regression is implemented widely in a large range of fields for comparing the influence of predictors, forecasting, and analyzing relationships. In multiple linear regression, each predictor factor is considered controlling for all other factors. The equation for this linear regression is:

$$LNMEDHVAL = \beta_0 + \beta_1 PCTVACANT + \beta_2 PCTSINGLES + \beta_3 PCTBACHMOR + \beta_4 LNNBELPOV100 + \varepsilon$$

Where:

- *LNMEDHVAL*: the logarithmic-transformed Median Household Value
- β_0 is the intercept term, which represents the value of the dependent variable when all the predictor variables are zero
- *PCTVACANT*: the percentage of vacant houses
- β_1 : the coefficient for *PCTVACANT*
- *PCTSINGLES*: the percentage of single house units
- β_2 : the coefficient for *PCTSINGLES*
- *PCTBACHMOR*: percentage of individuals with a bachelor's degrees or higher
- β_3 : the coefficient for *PCTBACHMOR*
- *LNNBELPOV100*: logarithmic-transformed number of households living in poverty
- β_4 : the coefficient for *LNNBELPOV100*
- ε is the error term where $\varepsilon \sim N(0, \sigma^2)$

Regression Assumptions

To get reliable and unbiased results from the regression model, five assumptions have to be met.

1. Linearity:

The relationship between the independent variables and the dependent variable should be linear and non-deterministic. This means that a change in the dependent variable should be proportional to the change in the independent variables. If this assumption is not met, the regression may not capture the true relationship of the variables, resulting in biased results. Linearity was checked by plotting the dependent variable against the predictor to visualize any clear non-linear patterns such as a curve or quadratic relationship between them. In general, the assumption is met if the relationship is approximately linear, but if there is no clear pattern, it is still best to fit a linear regression. If it is not linear, variable transformation or polynomial regression may be a better alternative. The assumption is satisfied if the plot shows a random scatter with a rather than a clear pattern.

2. Normality of residuals

The model residuals, the differences between the observed and predicted values, should follow a normal distribution. Normality of residuals is important for constructing confidence intervals, point estimation and hypothesis testing, with non-normality resulting in inaccurate results. To check for this assumption, a histogram of residuals can be plotted. Violations of normality of residuals may be due to the outcome and/or predictor variables being non-normal, and/or the linearity assumption being violated, but a non-linear transformation can address these issues. In certain situations, the problem with the residual distribution can be due to solely one or two huge errors, which should be inspected closely to check if they are not a result of data errors, whether they are logical, potentially re-occurring, and how influential to the model they are.

However, this is generally only an issue for smaller sample sizes as based on Central Limit Theorem, larger sample sizes of observations can help to minimize this issue.

3. Independence of observations

The observations in the dataset should also be independent of each other, meaning that one observation should not influence another. Violating this assumption could lead to inaccurate errors and inferences. An example of observations that are dependent on each other is time-series datasets and observations with spatial autocorrelation such as clustering.

4. Homoscedasticity

There should also be homoscedasticity, or constant variance of the residuals. This means that the spread of residuals should not be dependent or influenced by the independent variables. If heteroscedasticity is present, it means that the model is inadequately explaining the variability in the data, which can result in unreliable estimates. Heteroscedasticity can be addressed by transforming variables. To check for this assumption, the standardized residuals were plotted against the predicted values. Visually, there should not be a distinct pattern visible in the residuals as predicted values change.

5. No multicollinearity

Finally, independent variables should not exhibit strong correlations with one another. Multicollinearity arises when two or more predictor variables are highly correlated, complicating the ability to determine their individual impacts on the outcome variable. This can result in inflated standard errors and unreliable coefficient estimates. To assess this issue, the variance inflation factor (VIF) can be examined, where a VIF value exceeding 10 indicates significant multicollinearity. To mitigate this, highly correlated variables can either be removed or merged.

Parameters of Regression and Estimating Them

Parameters β_0 , β_1 and β_k were previously discussed under "Linear Regression". In multiple regression, there is no set number of variables that must be included, as each predictor variable has an associated coefficient. If there are k variables, the parameters will range from β_0 to β_k .

To estimate these parameters in multiple regression, given the number of observations and the number of predictors, the β hat coefficients from 0 to k are simultaneously selected to minimize the value of the Error Sum of Squares (SSE). The equation of SSE is as follows.

$$SSE = \sum_{i=1}^n (\varepsilon_i)^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \dots + \hat{\beta}_k x_{ki}) \right)^2 \quad (3)$$

Where:

- n : the number of observations
- ε_i : the residual or error term
- k : the number of predictors
- y_i : the actual value of the dependent variable for observation i
- $\hat{y}_i$: the predicted value of the dependent variable for observation i , based on the regression model
- $\hat{\beta}_0$: the estimated intercept of the regression model—the predicted value of y when all the predictors x are zero

- $\hat{\beta}_1, \hat{\beta}_2 \dots \hat{\beta}_k$: the estimated coefficients for the predictor variables $x_{1i}, x_{2i} \dots x_{ki}$ respectively. Each beta coefficient is the change in the y for a one-unit change in the corresponding x , holding other predictors constant.
- $x_{1i}, x_{2i} \dots x_{ki}$: the values of the predictor variables for observation i

The least squares estimates for the coefficients are obtained when the final value mentioned above is minimized.

Coefficient of Multiple Determination R^2

The coefficient of multiple determination, R^2 , quantifies the proportion of variance in the outcome variable that is accounted for by the predictor variables in a multiple regression model. It serves as an indicator of the model's overall fit to the data.

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST} \quad (4)$$

SSE (Sum of Squared Errors): Equation 3

SST (Total Sum of Squares):

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n \left(y_i - \frac{\sum_{i=1}^n (y_i)}{n} \right)^2 \quad (5)$$

Where:

- n : the number of observations
- y_i : the actual value of the dependent variable for observation i
- $\bar{y}$: the average value of y across all observations

And:

$$SSR \text{ (Regression Sum of Squares)} = SST - SSE \quad (5a)$$

The adjusted R^2 value is a modified version of the R^2 that takes into account the number of predictor variables in the model. It penalizes the inclusion of variables that do not enhance the model's fit, offering a more reliable measure of the model's explanatory power, especially when multiple independent variables are involved.

$$R_{adj}^2 = \frac{(n - 1)R^2 - k}{n - (k + 1)} \quad (6)$$

Where:

- R^2 : Equation 4
- n : the number of observations
- k : the number of independent variables

Hypothesis Testing

The goal of hypothesis testing is to use sample data to make inferences about the population from which the sample is from. An F-ratio, or a model utility test, is conducted before doing a hypothesis test. The F-ratio is used to test the overall significance of the regression model, by investigating if at least one or none of the independent variables is a significant predictor of the dependent variable.

The F-ratio is calculated as:

$$F = \frac{\frac{SSR}{k}}{\frac{SSE}{n-k-1}} = \frac{MSR}{MSE} \quad (7)$$

Where:

- *SSR: Sum of Squares Regression, Equation 5a*
- *SSE: Sum of Square of Errors, Equation 3*
- *n : the number of observations*
- *k : the number of predictors*
- *MSR: Mean Square Regression measures the average variance explained by the model*
- *MSE: Mean Square Error measures the average variance that is unexplained*

An F-ratio that is significantly larger than 1 indicates that the model explains a significant amount of variance in the outcome variable compared to the residuals. The hypotheses concerning the overall model fit is as follows.

$$H_0 : \beta_0 = \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$H_a : \beta_0 \neq 0 \text{ or } \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \text{ or } \beta_3 \neq 0 \text{ or } \beta_4 \neq 0$$

H_0 is the null hypotheses, while H_1 is the alternative hypothesis.

The hypotheses for each individual predictor variable is as follows.

$$H_0 : \beta_i = 0$$

$$H_a : \beta_i \neq 0$$

where $i = 1, 2, 3, 4$.

For each predictor variable, the goal is to determine whether there is a statistically significant relationship with the dependent variable, as stated in the alternative hypothesis (H_a).

If the p-value of the F-ratio is below 0.05, the null hypothesis (H_0) can be rejected at the 5%

significance level. If H_0 is not rejected, it implies that the predictor variable does not make a significant contribution to the model.

Additional Analysis

Stepwise regression is a basic data mining technique applied in multiple regression analysis to determine a subset of predictors that most enhance the model's predictive accuracy, guided by specific criteria such as p-values falling below a set threshold or the lowest Akaike Information Criterion (AIC) values, which assess the relative quality of statistical models. Nevertheless, this approach has certain drawbacks. First, it does not ensure an optimized final model, and there may be other models with comparable predictive capabilities that are not selected. Moreover, stepwise regression does not take into account the researcher's domain expertise. It is also crucial to recognize that this method might include irrelevant predictors while potentially omitting others that are truly significant.

K-fold Cross Validation

To assess the predictivity of our regression models, k-fold cross validation method, the resampling technique that evaluates the model's ability to generalize to unseen data by dividing the dataset into k subsets, were implemented. We used 5 – fold cross validation, by the CVlm function from the DAAG package in R. We applied 5-fold cross-validation to two regression models: full model included four predictor variables of PCTVACANT, PCTSINGLES, PCTBACHMOR, and LNNBELPOV100. Reduced Model, a simplified regression model included only PCTVACANT and MEDHHINC as predictors.

The each model is calculated with Root Mean Square Error (RMSE) metric, which measures the average magnitude of error between the predicted and actual values. RSME is calculated as the square root of the mean squared error (MRE).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

- y_i = *observed values*
- $\hat{y}_i$ = *predicted values*
- n = *number of observations*

Through this analysis, RMSE gives the guideline on the best version of the regression model with the best performance, with minimal prediction errors.

Results

Exploratory Results

The table below shows the summary of the following five statistics for the 1,720 block groups from the 2000 Philadelphia Census data after excluding outliers (see Methods): median house value (MEDHVAL), number of households in poverty (NBELPOV100), percentage of individuals with bachelor's degrees or higher (PCTBACHMOR), percentage of vacant houses (PCTVACANT), and percentage of single house units (PCTSINGLES).

Summary Statistic (2 decimal places)		
Variable	Mean	Standard Deviation
Dependent Variable		
Median House Value (\$)	66287.73	60006.08

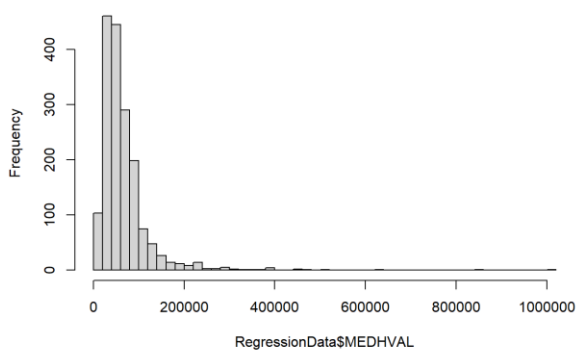
Predictors		
# of Households Living in Poverty	189.77	164.32
% of individuals with Bachelor's Degree or Higher	16.08	17.77
% of Vacant Houses	11.29	9.63
% of Single House Units	9.23	13.25

For the median house value, the standard deviation of \$60,006.08 nearly matches the mean value of \$66,287.73, indicating significant variability in house prices across the block groups. Furthermore, predictors such as the percentage of individuals with bachelor's degree or higher and the percentage of single house units see a standard deviation value higher than their respective mean values. This large variability suggests the data are likely spread out and may have skewed distributions.

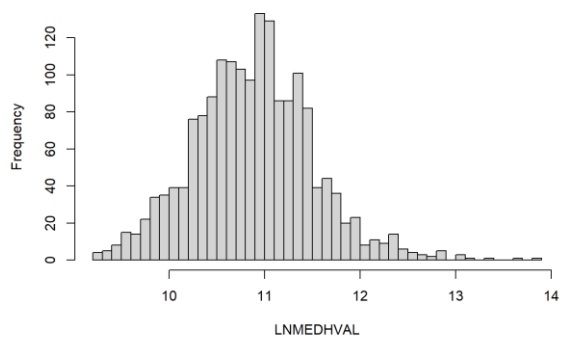
To explore this, the original distributions were compared to a logarithmic transformation of the dependent variable (MEDHVAL) the predictors. Specifically, a $\log(1 + \text{variable})$ transformation was used for the predictors with zero values, while the dependent variable was transformed using $\log(\text{variable})$. The distributions of the original and log-transformed variables are shown below.

Original Distribution	Logarithmic Transformation
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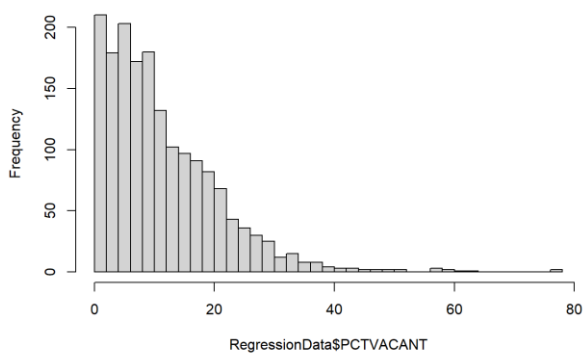
Histogram of RegressionData\$MEDHVAL



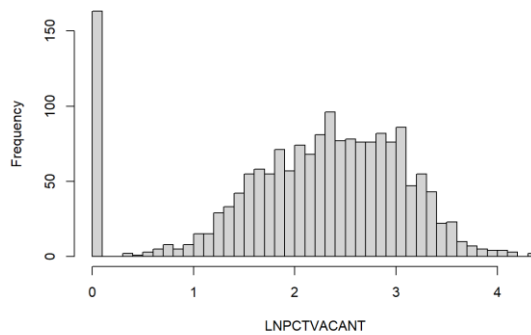
Histogram of LNMEDHVAL



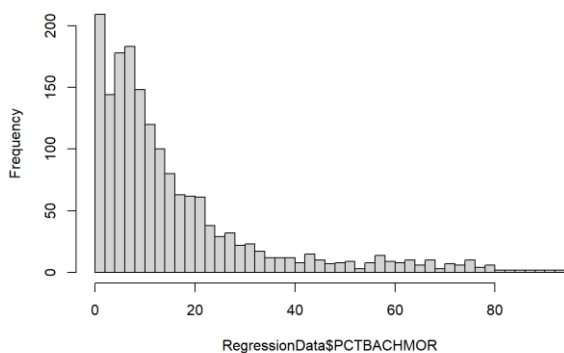
Histogram of RegressionData\$PCTVACANT



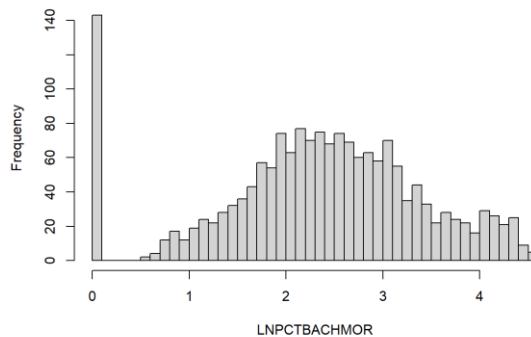
Histogram of LNPCTVACANT

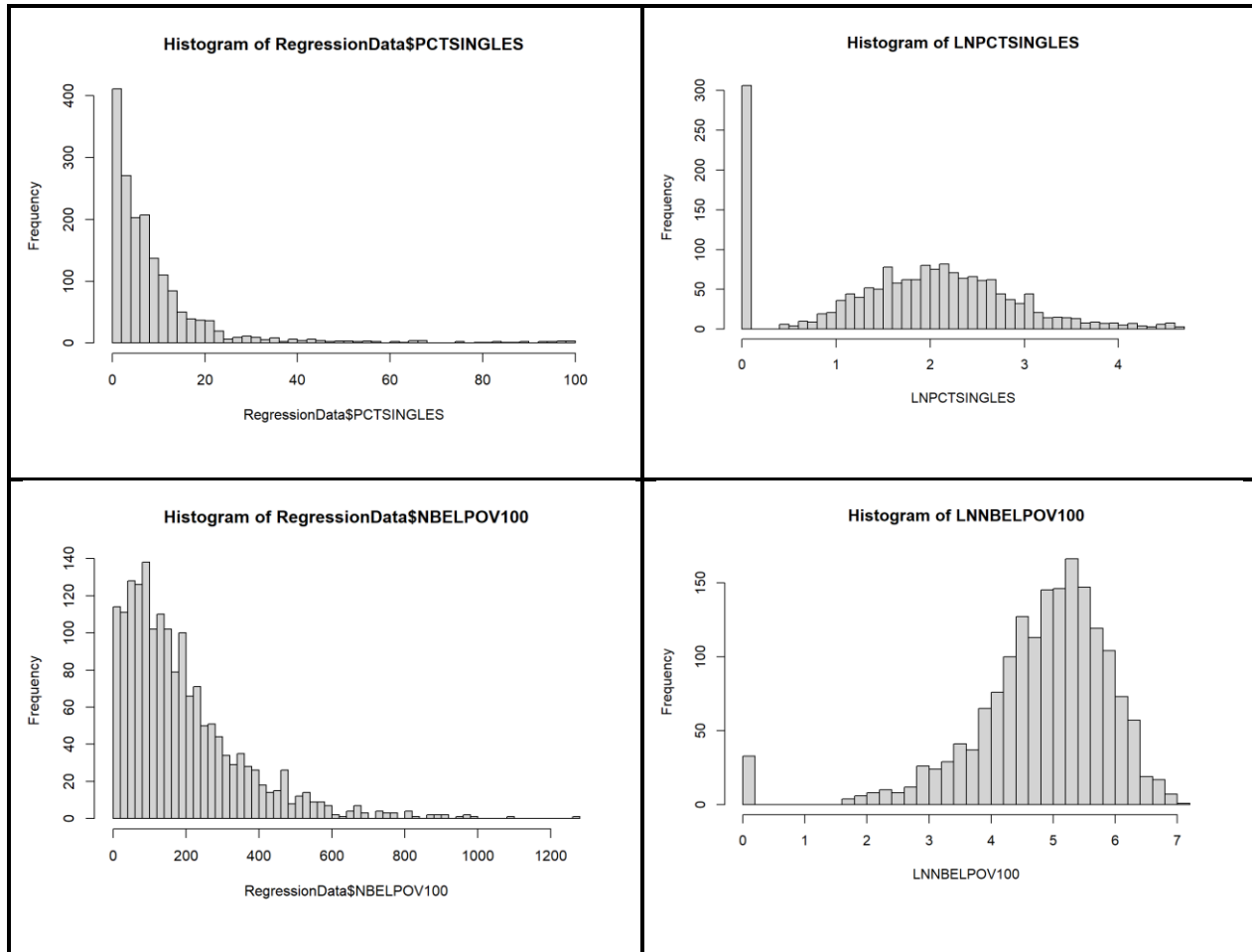


Histogram of RegressionData\$PCTBACHMOR



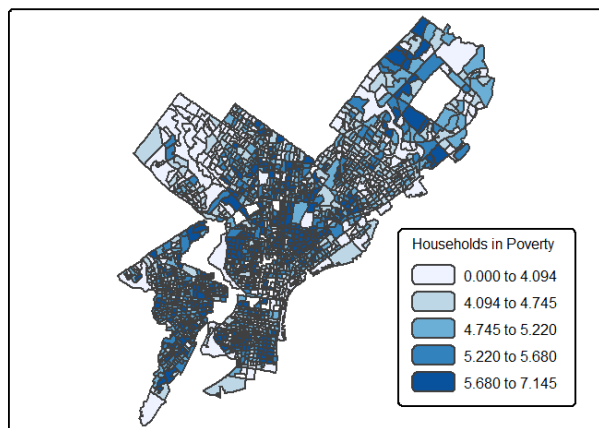
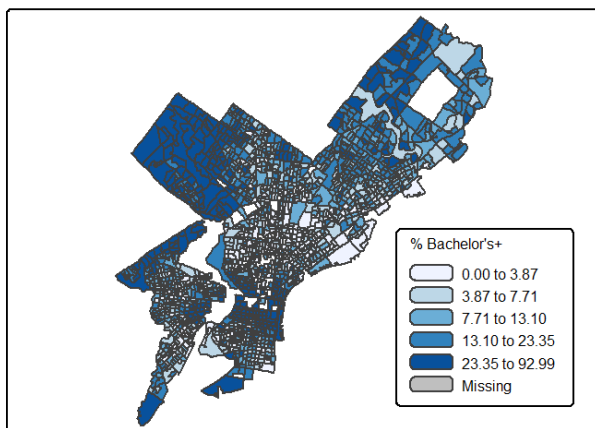
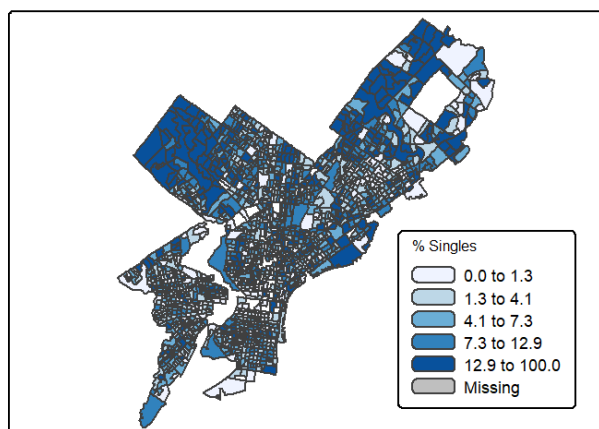
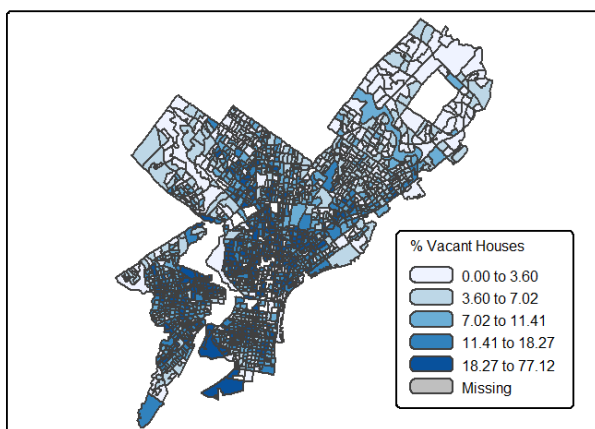
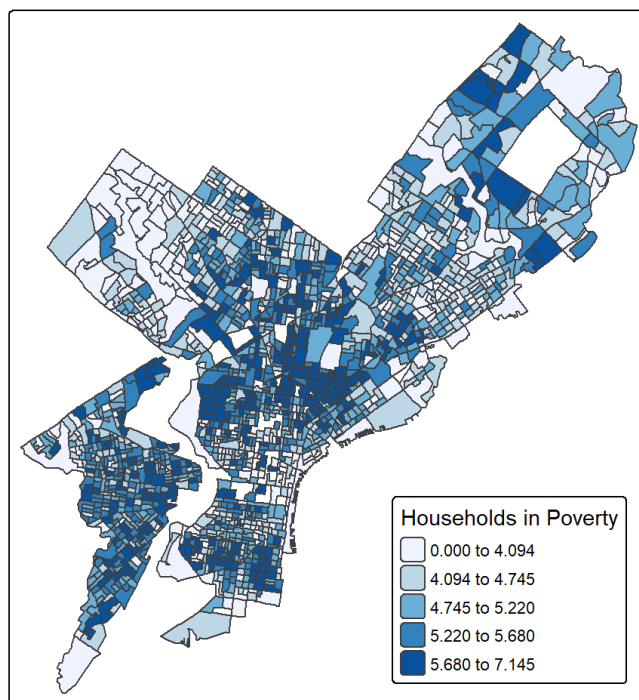
Histogram of LNPCTBACHMOR





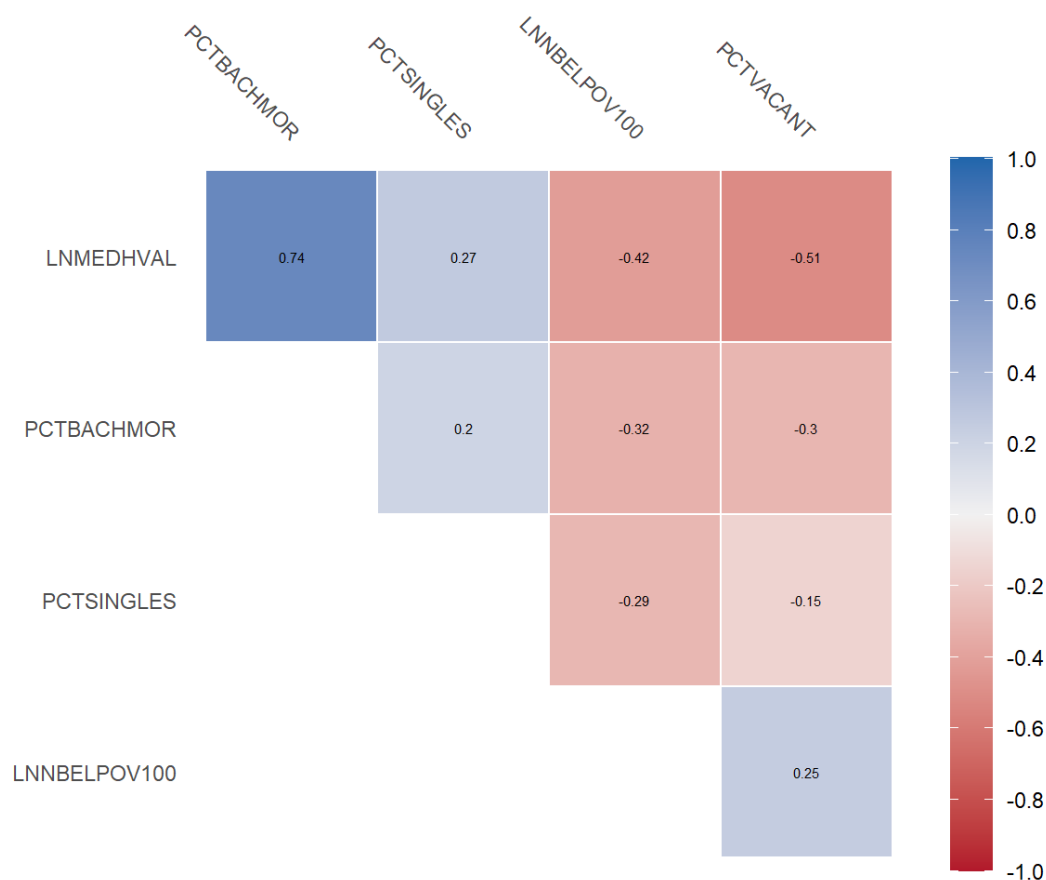
Looking at the histograms above, it's clear that none of the original variables closely follow a normal distribution. While the dependent variable (median house value) approaches a normal distribution following the log transformation, the predictor variables all exhibit clustering near zero following the log transformation. Despite this, the log transformed predictors still show improvement in normality. In particular, the log transformed NBELPOV100 closely resembles a normal distribution.

For the regression analysis, the log-transformed versions of the dependent variable (LNMEDHVAL) and NBELPOV100 will therefore be used alongside the original PCTBACHMOR, PCTSINGLES, and PCTVACANT. To observe the spatial distribution of each variable, the choropleth maps displayed in quantile breaks for each were examined.



Comparing the five maps, MEDHVAL, PCTBACHMOR, and PCTSINGLES have relatively similar distributions, while PCTVACANT and LNNBELPOV100 have more similar distributions. Based on the maps, it is expected that PCTBACHMOR will likely have a strong association with median house value. PCTVACANT is also likely to be highly negative correlated, with an inverse color pattern to the outcome variable. However, there is still some variation—percent below poverty shows areas of high poverty in the extreme northern area of Philadelphia, where the percent of vacant housing is not as high. While the maps do exhibit some similarities across the variables, maps alone do not allow us to confirm whether severe multicollinearity is present.

A correlation matrix of the predictors is thus plotted to further investigate potential multicollinearity.



Since the values of the correlation coefficients among PCTBACHMOR, PCTVACANT, PCTSINGLES, and NBELPOV100 are all less than 0.4, we can conclude that there is no severe multicollinearity among the predictors. The correlation matrix therefore supports our conclusions based on our visual comparison of predictor maps.

Regression Results

The group performed a regression analysis with the log-transformed median house value (LNMEDHVAL) as the dependent variable, and the log-transformed number households living in poverty (LNNBELPOV100), the percentage of individuals with bachelor's degrees or higher (PCTBACHMOR), the percentage of vacant houses (PCTVACANT), and the percentage of single house units (PCTSINGLES) as independent variables. Using the R command *lm(LNMEDHVAL ~ PCTVACANT + PCTSINGLES + PCTBACHMOR + LNNBELPOV100, data = data)*, we obtained the regression output and conducted an ANOVA analysis from R. The results indicate that PCTVACANT, PCTSINGLES, PCTBACHMOR, and ln LNNBELPOV100 are highly significant predictors of ln MEDHVAL ($p < 0.0001$ for all four variables).

```

Call:
lm(formula = LNMEDHVAL ~ PCTVACANT + PCTSINGLES + PCTBACHMOR +
    LNNBELPOV100, data = RegressionData)

Residuals:
    Min       1Q   Median       3Q      Max
-2.25825 -0.20391  0.03822  0.21744  2.24347

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.1137661  0.0465330  238.836 < 0.0000000000000002 ***
PCTVACANT    -0.0191569  0.0009779  -19.590 < 0.0000000000000002 ***
PCTSINGLES    0.0029769  0.0007032   4.234  0.0000242 ***
PCTBACHMOR    0.0209098  0.0005432  38.494 < 0.0000000000000002 ***
LNNBELPOV100 -0.0789054  0.0084569  -9.330 < 0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3665 on 1715 degrees of freedom
Multiple R-squared:  0.6623,    Adjusted R-squared:  0.6615
F-statistic: 840.9 on 4 and 1715 DF,  p-value: < 0.00000000000000022

Analysis of Variance Table

Response: LNMEDHVAL
      Df Sum Sq Mean Sq F value    Pr(>F)
PCTVACANT      1 180.392  180.392 1343.087 < 0.00000000000000022 ***
PCTSINGLES      1  24.543   24.543  182.734 < 0.00000000000000022 ***
PCTBACHMOR      1 235.118  235.118 1750.551 < 0.00000000000000022 ***
LNNBELPOV100    1  11.692   11.692   87.054 < 0.00000000000000022 ***
Residuals    1715 230.344    0.134
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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Mathematically, this regression can be expressed using the equation below.

$$\ln y = 11.114 - 0.0192x_1 + 0.00298x_2 + 0.0209x_3 - 0.0789 \ln x_4$$

where:

$$y = \text{MEDHVAL}$$

$$x_1 = \text{PCTVACANT}$$

$$x_2 = \text{PCTSINGLES}$$

$$x_3 = \text{PCTBACHMOR}$$

$$x_4 = \text{NBELPOV100}$$

$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ = coefficient of the independent variables with values inputted in

ϵ = known as an error term where $\epsilon \sim N(0, \sigma^2)$

We can analyze the results based on the regression output provided. First, a one percent increase in vacant housing (PCTVACANT) is associated with a $(e^{\beta_1} - 1) * 100\% = -1.90\%$ decrease in median house value. Conversely, a one percent increase in the share of single-house units (PCTSINGLES) corresponds to a $(e^{\beta_2} - 1) * 100\% = 0.298\%$ increase in median house value. A one percent rise in the percentage of individuals with a bachelor's degree (PCTBACHMOR) is linked to a $(e^{\beta_3} - 1) * 100\% = 2.11\%$ increase in median house value. Finally, a one percent increase in the number of households below the poverty level (LNNBELOWPOV100) is associated with a $(1.01^{\beta_4} - 1) \times 100\% = -0.0785\%$ change in median house value.

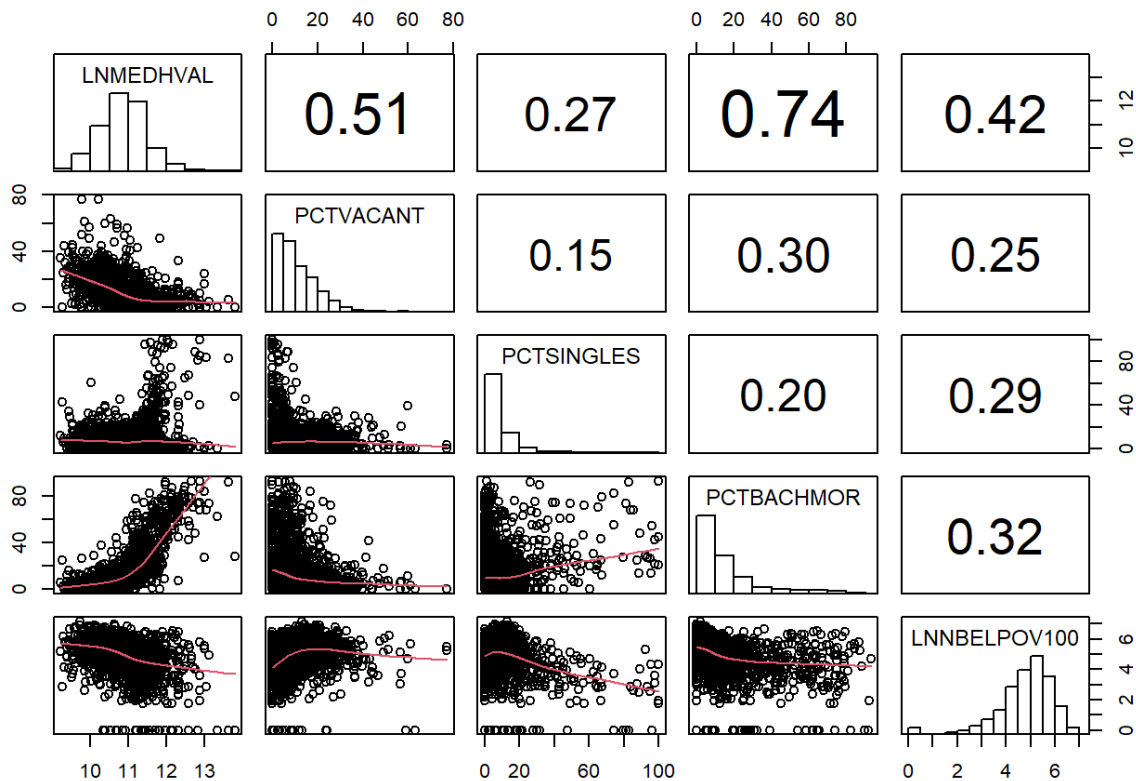
All of the predictors are statistically significant ($p < 0.0001$), indicating that PCTVACANT, PCTSINGLES, PCTBACHMOR, and LNNBELPOV100 are highly significant predictors of median house value. This means that if there were no true relationship between each predictor and the dependent variable (MEDHHINC), the probability of observing the estimated beta coefficients would be less than 0.0001. These extremely low probabilities allow us to confidently reject the null hypotheses ($H_0: \beta_1 = 0$, $H_0: \beta_2 = 0$, $H_0: \beta_3 = 0$, and $H_0: \beta_4 = 0$) in favor of the alternative hypotheses ($H_a: \beta_1 \neq 0$, $H_a: \beta_2 \neq 0$, $H_a: \beta_3 \neq 0$, and $H_a: \beta_4 \neq 0$) at conventional significance levels (α).

The R^2 value of 0.6622 indicates that 66.23% of the variance in median house value is explained by the predictors. The adjusted R^2 value of 0.6615 suggests that 66.15% of the variance is explained after accounting for the number of predictors (four in this case). The F-ratio of 840.9, with a p-value < 0.0001 , allows us to reject the null hypothesis that all β coefficients (for PCTVACANT, PCTSINGLES, PCTBACHMOR, and LNNBELPOV100) are zero.

The regression sum of squares (SSR), calculated as $SS(\text{PCTVACANT}) + SS(\text{PCTSINGLES}) + SS(\text{PCTBACHMOR}) + SS(\text{LNNBELPOV100}) = 180.392 + 24.543 + 235.118 + 11.692 = 451.745$, represents the total variance in the dependent variable explained by the model. The sum of squared errors (SSE) is 230.344, representing the unexplained variance in median house value. The addition of each predictor—PCTVACANT, PCTSINGLES, PCTBACHMOR, and LNNBELPOV100—reduces the SSE after controlling for the other variables in the model.

Regression Assumption Checks

Following the regression assumptions made, this section will check those assumptions to ensure that the five assumptions are met.

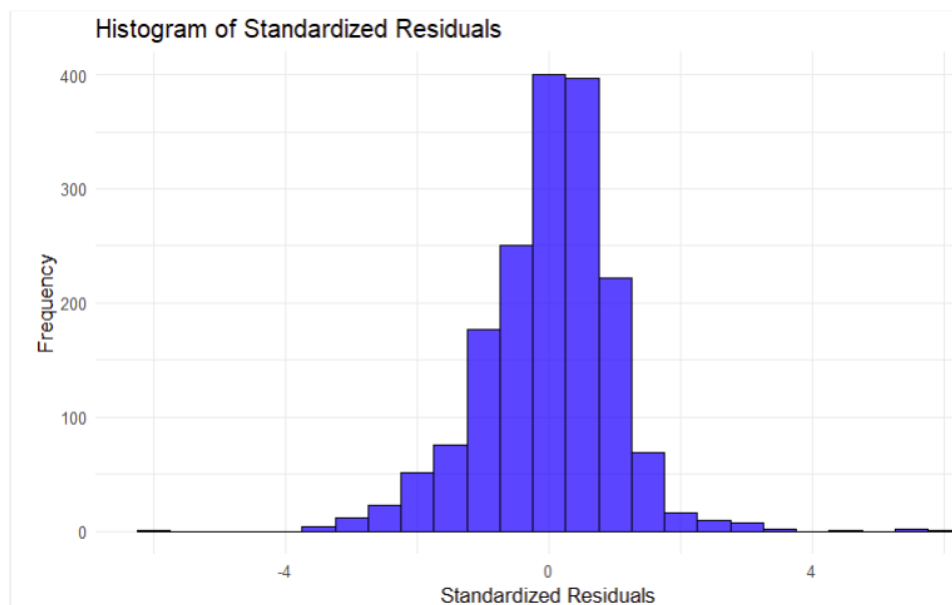


1. Linearity

Looking at the scatter plots of each predictor against the dependent variable of LNMEDHVAL reveals the relationship to not be linear. This is concerning as the non-linearity may reduce the accuracy of the results.

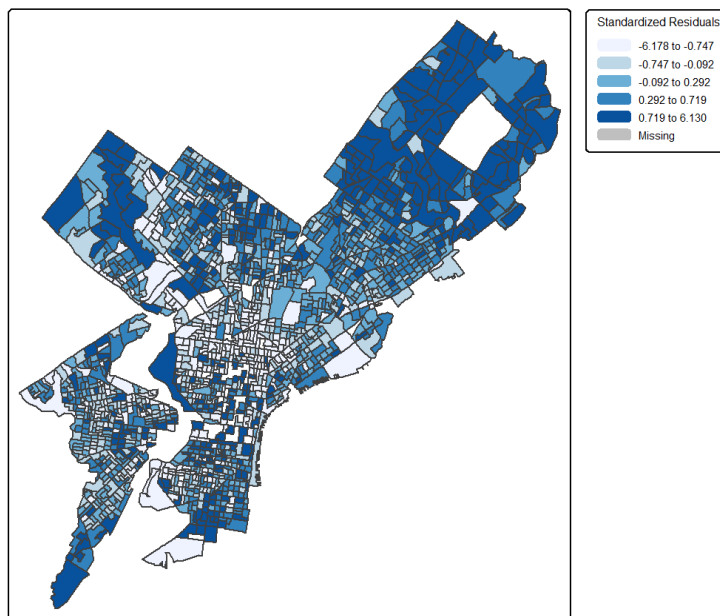
2. Normality of residuals

A histogram of the standardized residuals demonstrates a normal distribution, which shows that the residuals meet the normality assumption. Larger residuals serve as outliers, but they do not skew the data in any significant way. These outliers can suggest that our model does not perfectly depict particular neighborhoods due to other factors beyond our parameters.



3. Independence of observations

Choropleth Map of Standardized Residuals

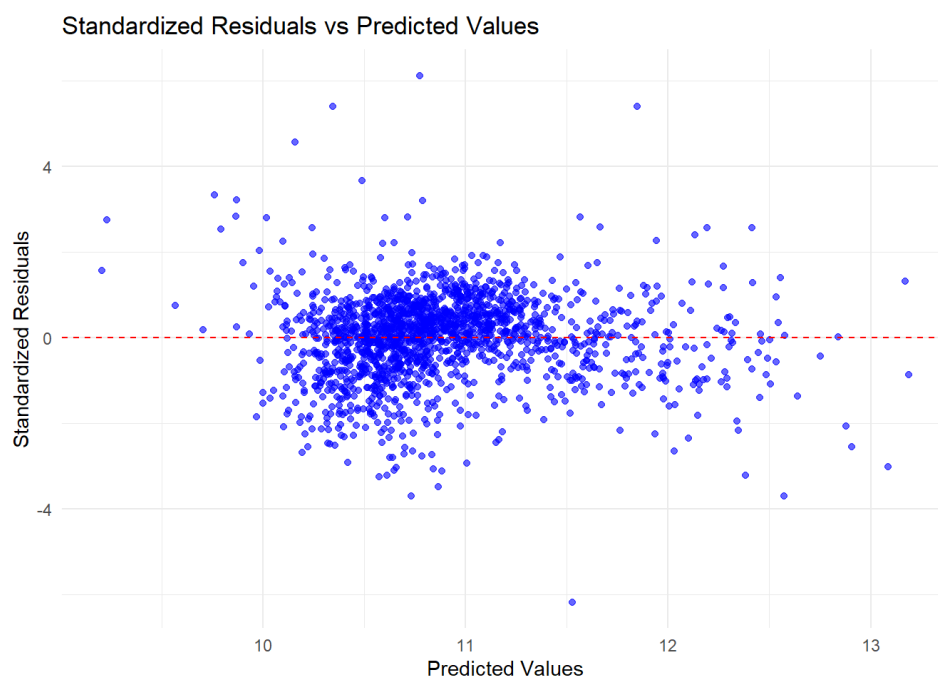


The assumption of independent observations in the Ordinary Least Squares (OLS) framework may be violated due to the spatial pattern of residuals. The choropleth map of standardized regression residuals reveals spatial clustering, with negative residuals concentrated in North and West Philadelphia. When considering the earlier maps of the dependent variable and predictors, there is evidence of spatial autocorrelation, suggesting that neighboring observations are not entirely independent. This lack of independence can result in biased standard errors, potentially affecting the validity of statistical inferences. Given these findings, spatial regression models may provide a more accurate representation of housing value determinants by explicitly accounting for spatial dependencies in the data.

4. Homoscedasticity

The standardized residuals appear to have a relatively constant variance across the predicted values, as seen in the residual's vs predicted values scatter plot. While there are a few outliers present, the overall pattern doesn't display a clear sign of heteroscedasticity. This

suggests that the variability in house prices is consistently explained by the predictors across the range of data.



5. No Multicollinearity

The correlation matrix shows there are no strong correlations among the independent variables. All r-values are below 0.4, suggesting that multicollinearity is not a concern in this model. All variance inflation factors (VIFs), which are all well below the commonly accepted threshold of 10, further support this claim. A lack of multicollinearity means the estimated coefficients are remain stable and reliable. Multicollinearity occurs when two or more predictors are highly correlated with each other, typically when the correlation coefficient exceeds 0.9 or falls below -0.9.

Additional Models

The root mean squared error (RMSE) analysis indicates that the full model outperforms the reduced model in predicting the log-transformed median home values (LNMEDHVAL). The

RMSE for the full model is 0.366, while the RMSE for the reduced model is 0.442. Also, the sum of squares (SS) and mean square (MS) values confirms that the full model better minimizes variance and prediction error, as the full model's MS is 0.134, while the reduced model's MS is 0.196. The full model with the smaller RMSE and MS demonstrates that the full model provides more accurate predictions with lower prediction errors.

The superior performance of the full model can be attributed to including all key predictors of PCTVACANT (vacancy rate), PCTSINGLES (proportion of single-family housing), PCTBACHMORE (percentage of residents with a bachelor's degree or higher), and LNNBELPOV100 (log-transformed count of households in poverty), while the reduced model only included the vacancy rate and MEDHHINC (median household income). Since the full model accounted for the additional predictors, it can be inferred that it led to a more comprehensive understanding of the factors influencing median home values in Philadelphia.

Full model

```
Sum of squares = 46.32   Mean square = 0.13   n = 344
overall (Sum over all 344 folds)
  ms
0.1342714
[1] "RMSE for the full model: 0.365942294678058"
```

Reduced model

```
Sum of squares = 76.79   Mean square = 0.22   n = 344
overall (Sum over all 344 folds)
  ms
0.1959939
[1] "RMSE for the reduced model: 0.44222719282485"
```

Final Results

```
[1] "RMSE for the full model: 0.365942294678058"
[1] "RMSE for the reduced model: 0.44222719282485"
[1] "the full model performs better."
```

However, while the full model performs better, it comes with increased complexity. If interpretability and simplicity are priorities, a slightly higher RMSE in the reduced model might still be acceptable. This means that given the notable RMSE difference of 0.366 and 0.442, dropping variable might result in a significant loss of predictive power.

Discussion and Limitations

This analysis applied ordinary least squares (OLS) regression to understand the factors that influence median house values across Philadelphia's census block groups. Four predictors were identified: the percentage of vacant houses (PCTVACANT), the percentage of single housing units (PCTSINGLES), the percentage of individuals with a bachelor's degree or higher (PCTBACHMOR), and the log-transformed number of households living in poverty (LNNBELPOV100). All four predictors were statistically significant ($p < 0.0001$). The percentage of individuals with a bachelor's degree demonstrated a strong positive relationship with house values, whereas the log-transformed number of households living in poverty had a negative effect. The significance of the percentage of vacant houses was expected, but negative impact on housing value highlights the detrimental effect of vacant properties on a neighborhood's appeal. Especially in Philadelphia, vacant properties are common and easy to identify around the city as the city's population has steadily decreased since 1950. Among all predictors, the percent of single units in a block group had the weakest influence on house values.

The model is strong demonstrated by an R^2 of 66.2%, indicating that the predictors explained a significant portion of the variance in median house values. The F-ratio test confirmed that the model effectively captured the variation in house prices. However, additional predictors could be used to strengthen this model such as access to public transportation, schools, proximity to parks, and how high the crime rate in a neighborhood is.

Stepwise Regression: The stepwise regression process retained all four predictors. This highlights their individual contributions to explaining house prices.

Cross-Validation: The root mean square error (RMSE) for the full 4-predictor model (10.894) was nearly identical to that of the reduced 2-predictor model (10.892), which included PCTVACANT and MEDHHINCOME. This shows that MEDHHINCOME is a powerful

predictor, although retaining all predictors provides a more encompassing understanding of the factors influencing house values.

Model Limitations

The model used in this study has some limitations that must be address. An assumption of linearity is not strongly satisfied, which lowers the accuracy and reliability of the results of the regression. Secondly, spatial autocorrelation in the residuals indicate that nearby observations may not be independent across the city, which violates a key assumption of OLS regression. Lastly, using the raw number of households living in poverty is not an ideal predictor since it does not account for population differences across block groups where a percentage like the other categories are more appropriate.

Ridge or LASSO Regression

Ridge and LASSO regression are regularization methods aimed at mitigating multicollinearity and preventing overfitting in models with numerous predictors. Ridge regression reduces the magnitude of coefficients, while LASSO has the added capability of shrinking some coefficients to zero, effectively performing variable selection. In this study, since multicollinearity was not a major concern and the model used only four predictors, using either Ridge or LASSO regression is unnecessary. However, if the model had more predictors, these techniques could prove valuable in controlling overfitting and identifying key predictors.

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